EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

BIOMOLECULES

1. The nitrogenous base having two possible hydrogen bonding sites is –

(1) Thymine (2) Cytosine

(3) Guanine (4) None

BM0001

2. The correct statement in respect of protein haemoglobin is that it –

(1) Maintains blood sugar level

(2) Acts as an oxygen carrier in the blood

(3) Forms antibodies and offers resistance to diseases

(4) acts as a catalyst for biological reactions

BM0002

3. The hormone that helps in the conversion of glucose to glycogen is –

(1) Bile acids (2) Adrenaline

(3) Insulin (4) Cortisone

BM0004

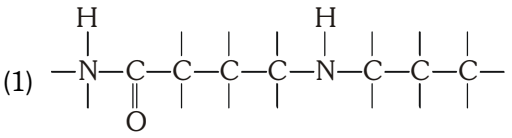
4. The helical structure of protein is stabilized by –

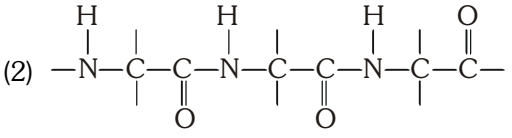
(1) Hydrogen bonds (2) ether bonds

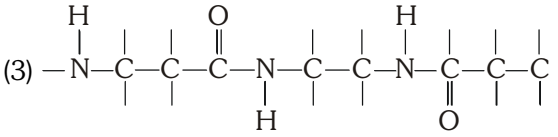
(3) peptide bonds (4) dipeptide bonds

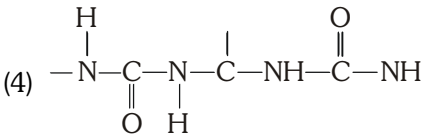
BM0005

5. Which of the following structures represents the peptide chain –

(1) 

(2) 

(3) 

(4) 

BM0007

6. Which functional group participates in disulphide bond formation in proteins –

(1) Thioether (2) Thiol

(3) Thioester (4) Thiolactone

BM0008

7. The cell membranes are mainly composed of –

(1) Phospholipids (2) Proteins

(3) Fats (4) Carbohydrates

BM0009

8. Which is simplest amino acid –

(1) Alanine (2) Asparagine

(3) Glycine (4) Tyrosine

BM0010

9. Which of the following biomolecules is insoluble in water –

(1) α–Keratin (2) Haemoglobin

(3) Ribonuclease (4) Adenine

BM0011

10. The presence or absence of hydroxy group on which carbon atom of sugar differentiates RNA and DNA ?

(1) 3rd (2) 4th (3) 1st (4) 2nd

BM0013

11. The change in the optical rotation of freshly prepared solution of glucose is known as :-

(1) tautomerism (2) racemisation

(3) specific rotation (4) mutarotation

BM0014

12. Which one of the following bases is not present in DNA ?

(1) Cytosine (2) Thymine

(3) Quinoline (4) Adenine

BM0016

13. Lysine; $H_2N-(CH_2)_4-\underset{\underset{NH_2}{|}}{CH}-COOH$ is :-

(1) α-Amino acid

(2) γ-amino acid

(3) Amino acid synthesised

(4) β-Amino acid

BM0017

14. In fibrous protein, polypeptide chains are held toegether by :-
(1) Vander waals forces
(2) Disulphide linkage
(3) Electrostatic forces attraction
(4) None of these
BM0018

15. Which is correct in following –
(1) Monosaccharides also known as sugars
(2) Polysaccharides are non sugars
(3) Maltose and Lactose are reducing sugar
(4) All of these
BM0019

16. Structure of glycogen is similar to :-
(1) Alylose (2) Amylopectin
(3) Cellulose (4) Glucose
BM0020

17. Which of the following gives osazone different from the other three :-
(1) Glucose (2) Mannose
(3) Galactose (4) Fructose
BM0021

18. Anomers of glucose (α-form & β-form) are differ in the stereochemistry at which carbon –
(1) C-1 (2) C-2
(3) C-3 (4) All of these
BM0022

19. Sucrose in presence of invertase on hydrolysis gave –
(1) Glucose (2) Fructose
(3 Ethyl alcohol (4) 1 & 2 both
BM0023

20. Which of the following B group vitamins can be stored in our body.
(1) Vitamin B_1 (2) Vitamin B_2
(3) Vitamin B_6 (4) Vitamin B_{12}
BM0024

21. Which of the following are polysaccharides –
(a) Starch (b) cellulose
(c) dextrins (d) glycogen
(1) a, b & c (2) a, b, d
(3) a & c (4) a,b, c, d
BM0025

22. Cellulose can not be tested by followings –
(1) Fehling's solution (2) Tollen's reagent
(3) Both of these (4) None of these
BM0026

23. On hydrolysis of proteins, the product is/are –
(1) Amino acids (2) Peptides
(3) Enzymes (4) 1 & 2 both
BM0027

24. Which of the following do not have hemiacetal group:-
(1) Fructose (2) Maltose
(3) Sucrose (4) Glucose
BM0028

25. In amino acids, more number of amino than carboxyl groups makes it –
(1) acidic (2) Basic
(3) Neutral (4) None of these
BM0029

26. Which amino acids are called non essential –
(1) those which can be synthesized in the body.
(2) those which have more amino groups as compared to carboxyl groups
(3) those which have equal number of amino acid and carboxyl groups
(4) None of these
BM0030

27. Which of the following is not essential amino acid –
(1) Serine (2) Lysine
(3) Threonine (4) Tryptophan
BM0031

28. In acidic & alkaline solution amino acids exists as a –
(1) Positive ion & negative ion respectively
(2) Negative ion & positive ions respectively
(3) Neutral in both medium
(4) None of these
BM0032

29. In which of following shapes are found in tertiary structure of proteins –
(1) Fibrous (2) Globular
(3) Both of these (4) None of these
BM0033

30. The example of globular protein is
(1) Silk (2) Collagen
(3) Haemoglobin (4) All of these
BM0034

31. If a native protein is subjected to physical or chemical treatment which may disrupt its various forms without affecting its primary structure, are called –
(1) Inactive protein
(2) Denatured protein
(3) Both of these
(4) None of these
BM0035

32. The coagulation of egg protein while boiling of egg is called –
(1) Reversible protein denaturation
(2) Irreversible protein denaturation
(3) Renaturation
(4) None of these
BM0036

33. An enzyme molecule may contain protein and non-protein part. Non-protein part is known as:-
(1) Holoenzyme
(2) Cofactor
(3) Inverted enzyme
(4) None of these
BM0037

34. Cofactor which gets attached to the enzyme at the time of reaction are known as
(1) Coenzymes (2) Apoenzyme
(3) Prothetic group (4) None of these
BM0038

35. Mainly DNA is localized in –
(1) Cytoplasm (2) Nucleus
(3) Mitochondria (4) Chloroplasts
BM0039

36. RNA contains following pyrimidine bases –
(a) Thymine (b) Uracil
(c) Cytosine (d) Adenine
(1) b & c (2) a, b, c
(3) a, b, d (4) All of these
BM0040

37. DNA molecules can duplicate themselves-called –
(1) Replication (2) Translation
(3) Transcription (4) None of these
BM0041

38. Which of the following carbohydrates are branched polymer of glucose.
(1) Glycogen (2) Amylopectin
(3) Cellulose (4) Both (1) & (2)
BM0042

39. Vitamin K
(1) Is phylloquinone
(2) Soluble in oils and fats
(3) Deficiency lenthens the blood clotting
(4) All of these
BM0043

40. Which of the following is not an amino acid
(1) Histidine (2) Benzidine
(3) Alanine (4) Proline
BM0044

41. Vitamin C is called :
(1) Antisterility (2) Antiscurvy
(3) both of these (4) None of these
BM0045

42. Reducing sugars are one which :
(1) reduce fehling's solutions
(2) not reduce tollen's reagent
(3) Have bonded aldehydic or ketonic groups
(4) All of these
BM0046

43. Multiple deficiencies caused by lack of more than one vitamin are more common in human beings. This condition of vitamin deficiency is known as
(1) avitaminosis
(2) xerophthalmia
(3) convulsions
(4) None of these
BM0047

44. Which vitamin is synthesize in human body from carotene @neetquestionpaper
(1) Vitamin-A
(2) Vitamin-C
(3) Vitamin-K
(4) All of these
BM0048

45. Which of the following is correct about H–bonding in nucleotide –
(1) A–T G–C
(2) A–G T–C
(3) G–T A–C
(4) A–A T–T
BM0049

Z:\NODE02\B080-BA\TARGET\CHEM\ENG\MODULE-6\04 BIOMOLECULES\02-EXE P65

POLYMER

46. Which of the following is monomer unit of polystyrene :-

(1) $-CH_2-\underset{\underset{CN}{|}}{CH}-$ (2) $-CF_2-CF_2-$

(3) $-CH_2-\underset{\underset{Cl}{|}}{CH}-$ (4) $-\underset{\underset{C_6H_5}{|}}{CH}-CH_2-$

PL0050

47. Weakest intermolecular forces are present in :-

(1) Neoprene (2) Terylene

(3) Polystyrene (4) Bakelite

PL0051

48. Thermosetting polymer, Bakelite is formed by the reaction of phenol with :-

(1) CH_3CH_2CHO (2) CH_3CHO

(3) HCHO (4) HCOOH

PL0052

49. Which one is classified as a condensation polymer?

(1) Teflon (2) Acrylonitrile

(3) Dacron (4) Neoprene

PL0053

50. Novolac is a :

(1) linear polymer of urea and formaldehyde

(2) crosslink polymer of urea and formaldehyde

(3) linear polymer of phenol and formaldehyde

(4) crosslink polymer of phenol and formaldehyde

PL0054

51. Which of the following is not a semisynthetic polymer.

(1) cis-polyisoprene (2) cellulose nitrate

(3) cellulose acetate (4) valcanised rubber

PL0055

52. Branched chain polymer is-

(1) High density polythene

(2) Low density polythene

(3) Poly vinyl chloride

(4) Bakelite

PL0189

53. Mechanical properties like tensile strength, elasticity, toughness etc, are governed by intermolecular forces. These intermolecular forces are -

(1) Hydrogen bond

(2) Vander waals force

(3) Both of these

(4) None of these

PL0190

54. Monomer unit of Nylon-6,6 is/are -

(1) Adipic acid

(2) 1,4-butamethylene diamine

(3) Caprolactum

(4) All of these

PL0191

55. Which of following polymer is used as glass reinforcing materials in safety helmets-

(1) Terylene (2) Teflon

(3) Terephthalic acid (4) Ethylene glycol

PL0192

56. Polymer used in radio and television cabinets-

(1) PVC (2) *Poly styrene*

(3) *Poly propene* (4) Glyptal

PL0193

CHEMISTRY IN EVERYDAY LIFE

57. Aspirin is also known as –

(1) Methyl salicyclic acid (2) acetyl salicylate

(3) Methyl salicylate (4) Acetyl salicylic acid

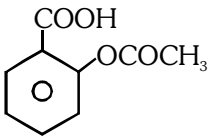
CE0056

58. Paracetamol is :-

(1) Analgesic (2) Antipyretic

(3) Both (4) Antiseptic

CE0057

59.  is used as

(1) Antiseptic (2) Analgesic

(3) Antibiotic (4) Micro organism

CE0058

60. Which among the following is a tranquilizer ?

(1) Equanil (2) promethazine

(3) Omeprazole (4) Cimetidine

CC0059

61. Which is incorrect statement ?

(1) Drugs are chemicals of low molecular masses (100-500 μ)

(2) When drugs interact with macromolecular target and produce a biological responce, they are called as potential poisons

(3) Use of chemicals for threrapeutic effect is called chemotherapy

(4) Medicines are used in diagnosis, prevention and treatment of diseases

CE0194

62. Drugs which interfere with natural action of histamine by competing with histamine are called.
(1) Antidepressant (2) Antihistamine
(3) Antimicrobial (4) Antipyretic
CE0195

63. Barbituric acid and its derivative constitutes an important class of
(1) Narcotic analgesic (2) Antiallergic
(3) Tranquilizers (4) Antimicrobial
CE0196

64. Drugs bind to a site (other than enzyme's active site). This site is named is.
(1) Competitive site (2) Allosteric site
(3) Non-active site (4) Super active site
CE0197

65. Name of drug that bind to the receptor site and inhibit its natural function
(1) Antagonist
(2) Antireceptor
(3) Antacid
(4) Non-additive analgesic
CE0198

66. First popular antacid drug was
(1) Cimetidine (2) Ranitidine
(3) Zantac (4) Equanil
CE0199

67. Which chemical is responsible for mood change.
(1) Noradrenaline (2) Dimetapp
(3) Seldane (4) All
CE0200

68. Which is not an example of tranquilizer
(1) Iproniazid (2) Phenelzine
(3) Ofloxacin (4) Barbiturates
CE0201

69. Morphine is used in
(1) Cardiac pain (2) Terminal cancer
(3) Child birth (4) All
CE0202

70. The first effective antibacterial agent discovered by Ehrlich
(1) Prontosil (2) Azo dye
(3) Salvarsan (4) Penicillin
CE0203

71. Which is not Bactericidal antibiotic.
(1) Penicillin (2) Aminoglycosides
(3) Ofloxine (4) Erythromycine
CE0204

72. Which is not Bacteriostatic antibiotic.
(1) Erythromycin (2) Penicillin
(3) Tetracycline (4) Chloromphenicol
CE0205

73. Which broad spectrum antibiotic is given orally in case of typhoid and dysentery.
(1) Ampicillin (2) Amoxycilline
(3) Ofloxacin (4) Chloromphenicol
CE0206

74. Find incorrect match

Drug	**Category**
(1) Chlordiazepoxide	Tranquilizer
(2) Chloramphenicol	Antibiotic
(3) Veronal	Antiseptic
(4) Sulphanilamide	Antibiotic

CE0207

75. Find incorrect match

Drug	**Treatment of**
(1) Salvarsan	Syphilis
(2) Chloramphenicol	Typhoid
(3) Morphine	Cardiac pain
(4) Mixture of $Mg(OH)_2$ and $Al(OH)_3$	Analgesic

CE0208

76. Which artificial sweeterner is of great value to diabetic person.
(1) Aspartame (2) Saccharin
(3) Sucrose (4) Alitame
CE0209

77. Which is mild tranquilizer used in controlling depression and hypertension.
(1) Meprobamate (2) Norethindrone
(3) Ethynylestradiol (4) Terpineol
CE0210

78. 0.2% solution of phenol and 0.2-0.4 ppm chlorine in aqueous solution respectively behave as.
(1) Antiseptic, Disinfectant
(2) Disinfectant, Antiseptic
(3) Disinfectant, Antacid
(4) Antiseptic, Antacid
CE0211

79. The first popular artificial sweetening agent is.
(1) Saccharin (2) Asparfame
(3) Alitame (4) Sucralose
CE0212

80. Which is stable at cooking temperature & doesn't provide calories.
(1) Aspartame (2) Saccharin
(3) Sucralose (4) Alitame
CE0213

81. Highest sweetness value found in
(1) Aspartame (2) Alitame
(3) Saccharin (4) Sucralose
CE0214

82. Which sweetening agent is used in soft drinks.
(1) Aspartame (2) Saccharin
(3) Sucralose (4) Alitame
CE0215

83. Which is unstable at cooking temperature
(1) Aspartame (2) Alitame
(3) Saccharin (4) All
CE0216

84. Soap are sodium or potassium salt of fatty acid unlike.
(1) Palmitic acid (2) Oleic acid
(3) 1 & 2 (4) Maleic acid
CE0217

85. Which enhance lathering property of soap.
(1) Sodium carbonate
(2) Sodium rosinate
(3) Sodium stearate
(4) Trisodium phosphate
CE0218

86. To make soft soap, saponification contain.
(1) NaOH (2) KOH
(3) Glycesol (4) Glycol
CE0219

87. Hair conditioners contain which kind of detergent.
(1) Cationic (2) Anionic
(3) Non ionic (4) All
CE0220

88. Which is used as food preservatives
(1) NaCl (2) Sugar
(3) Vegetable oil (4) All
CE0221

89. Which is added to soap to impart antiseptic properties.
(1) Bithionol (2) Soframycin
(3) Glycol (4) Ethanol
CE0222

EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**1**	**4**	**3**	**1**	**2**	**2**	**1**	**3**	**1**	**4**	**4**	**3**	**1**	**2**	**4**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**3**	**1**	**4**	**4**	**4**	**3**	**4**	**3**	**2**	**1**	**1**	**1**	**3**	**3**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**2**	**2**	**2**	**1**	**2**	**1**	**1**	**4**	**4**	**2**	**2**	**1**	**1**	**1**	**1**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**4**	**1**	**3**	**3**	**3**	**1**	**2**	**3**	**1**	**1**	**2**	**4**	**3**	**2**	**1**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**2**	**2**	**3**	**2**	**1**	**1**	**1**	**3**	**4**	**1**	**4**	**2**	**4**	**3**	**4**
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	
Ans.	**2**	**1**	**1**	**1**	**3**	**2**	**1**	**1**	**4**	**2**	**2**	**1**	**4**	**1**	

EXERCISE-III (Analytical Questions) — Master Your Understanding

1. In which of the following properties of an open chain structure of glucose could not be explained by Baeyer –
(1) Glucose contain aldehyde group, but does not give schiff's test
(2) Glucose does not react with sodium hydrogen sulphite and ammonia
(3) The pentaacetate of glucose does not react with hydroxy amine indicates absence of –CHO group
(4) All of these

BM0139

2. When glucose was crystallized from a concentrated solution at 30°C & 98°C. It gave α & β form of glucose respectively what are the angle of rotation $[\alpha_D]$ for both forms –
(1) (+) 111° & (+)19.2° (2) (+)19.2° & (–) 111°
(3) (+) 111° & (–)19.2° (4) (–) 111° & (–)19.2°

BM0140

3. Dinucleotide is obtained by joining two nucleotides together by phosphodiester linkage. Between which carbon atoms of pentose sugar of nucleotides are these linkages presents.
(1) 5' and 3' (2) 1' and 5'
(3) 5' and 5' (4) 3' and 3'

BM0141

4. Which statement is incorrect in following –
(1) Sucrose is dextrorotatory
(2) After hydrolysis of sucrose, dextrorotatory glucose and laevorotatory fructose will obtained
(3) Laevorotation of fructose is more than dextrorotation of glucose
(4) Cane sugar gives nonequimolar mixture of D-(+)-glucose and D-(–)-fructose

BM0143

5. On hydrolysis of one mole of maltose two moles of D-glucose are obtained. These two glucose units are linked together through a α-glycoside linkage between –
(1) C-2 of one unit and C-4 of another unit
(2) C-1 of one unit and C-2 of another unit
(3) C-1 of one unit and C-4 of another unit
(4) C-2 of one unit and C-3 of another unit

BM0145

6. Which statement is/are incorrect –
(1) Natural starch has approximately 10-20% of amylose and 80-90% of amylopectin.
(2) Amylose is water soluble and gives blue colour with iodine
(3) Amylopectin is a branched chain polysaccharides insoluble in water and does not give blue colour with iodine
(4) Starch is not hydrolysed by enzyme amylase present in saliva

BM0146

7. Which of the following statement is true for proteins –
(1) Proteins are high molecular mass complex biopolymers of amino acids present in all living cells
(2) Protein may contains phosphorous & iodine
(3) Proteins may contains iron, copper, zinc and manganese
(4) All of these

BM0147

8. Which of following are essential amino acids –
(a) Aspartic acid (b) Leucine
(c) Valine (d) Glycine
(e) Alanine
(1) a, b & c (2) a, b, d & e
(3) b & c (4) a, c, d & e

BM0148

9. At a certain hydrogen ion concentration (pH), the dipolar ion exists as a neutral ion and does not migrate to either electrode. What is the name of pH point for amino acids –
(1) isoelectric point (2) isoprotonic point
(3) isotonic point (4) None of these

BM0149

10. Which of the following reactions of glucose can be explained only by its cyclic structure.
(1) Glucose forms penta acetate
(2) Glucose reacts with hydroxylamine to form an oxime
(3) Penta acetate of glucose does not react with hydroxylamine
(4) Glucose is oxidised by nitric acid to gluconic acid

BM0150

11. Protein found in biological system with definite configuration and biological activity is called –
(1) Amino acids (2) Native protein
(3) Conformer protein (4) Inactive protein
BM0151

12. Which of following statement is not true for enzyme –
(1) Some enzymes can be non proteins also
(2) Prothetic groups which get attached to enzyme at the time of reaction are known as cofactor.
(3) Enzymes provides lower activation pathways there by increasing the rate of reaction
(4) None of these
BM0153

13. Which of following statements is incorrect –
(1) Proteins have a polyamide chain while nucleic acids contains a poly phosphate ester chain
(2) Nucleic acids are long chain polymers of nucleotides
(3) RNA is major source of genetic information which is copied into a DNA molecules
(4) Proteins are synthesized in a process involving translation of RNA.
BM0154

14. Complete hydrolysis of DNA or RNA yields following–
(1) Ribose in RNA & deoxyribose in DNA
(2) Hetrocyclic nitrogenous purines base
(3) Hetrocyclic nitrogenous pyrimidines
(4) All of these
BM0155

15. Nucleosides are –
(1) Base + sugar = Nucleoside
(2) N–glycosides of purine or pyrimidine bases with pentose sugar
(3) Both of these
(4) None of these
BM0157

16. RNA molecules in turn directs the synthesis of specific proteins which are characteristic of each kind of organism. This process is known as –
(1) Translation (2) Transcription
(3) Replication (4) None of these
BM0158

17. A single stand of DNA can act as a template on which a molecule of RNA is synthesized in a specific manner. This process is known as –
(1) Replication (2) Transcription
(3) Translation (4) None of these
BM0159

18. Which is correct statement –
(1) Starch is polymer of α–glucose
(2) Amylose is a component of cellulose
(3) Protiens are composed of only one type of amino acid
(4) In cyclic structure of fructose, there are four carbons and one oxygen atom
BM0162

19. Which is incorrect about histamine.
(1) Histamine stimulates the secretion of pepsin & HCl in the stomach
(2) Histamine is a potent vasodilator
(3) Histamine is also responsible for nasal congestion associated with cold or allergic response to pollen.
(4) Histamine doesn't affact muscles of the bronchi
BM0230

20. Brompheniramine & Terfenadine drugs belongs to which category.
(1) Antacid (2) Antihistamine
(3) Transquilizer (4) Antiseptic
CE0231

21. Find incorrect statement :
(1) Tranquilizer & analgesis are neurologically active drugs
(2) Barbiturates are sleep producing agents
(3) Tranquilizer are used to reduce stress, mental diseases
(4) Analgesis are used to reduce pain with causing impairement of consciousness
CE0232

22. Find incorrect statement
(1) Detergent involves soap and synthetic detergents
(2) Sodium soap and potassium soap are soluble in water
(3) $C_{17}H_{35}COONa$ is sodium stearate and used as a soap
(4) During saponification, soap is obtained in solid form
CE0233

23. Which of the following is/are condensation polymers-
(1) Dacron (2) Nylon 6
(3) Nylon 6,6 (4) All of these
PL0234

24. Free radical generating initiator (catalyst) is/are :
(1) Benzoyl peroxide
(2) Acetyl peroxide
(3) Tert-butyl peroxide
(4) All of these
PL0235

25. Novolac on heating with formaldehyde *under goes* cross linking to form an infusible solid mass is called.
(1) Bakelite
(2) Malamine-formaldehyde
(3) Terylene
(4) Nylon-6
PL0236

26. Correct statements about vulcanisation of rubber is/are-
(1) Rubber gets stiffened
(2) In this process, raw rubber is heated with sulphur and appropirate additive at temperature range 373 K to 415 K
(3) By this process physical properties of rubber improve
(4) All of these
PL0237

27. Arrange following polymers in increasing order of their intermolecular force-
(i) Nylon-6
(ii) Neoprene
(iii) Poly vinyl chloride
(1) i < ii < iii (2) ii < iii < i
(3) iii < i < ii (4) iii < ii < i
PL0238

28. Biodegradable polymer is-
(1) PHBV (2) Nylon 2 - Nylon 6
(3) Both of these (4) None of these
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EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**1**	**1**	**4**	**3**	**4**	**4**	**3**	**1**	**3**	**2**	**2**	**3**	**4**	**3**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	**1**	**2**	**1**	**4**	**2**	**4**	**4**	**4**	**4**	**1**	**4**	**2**	**3**		